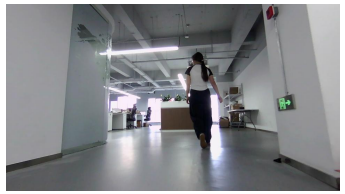
(a) DobotTrack VLA

Track the person with <description>

Text embedding



Past observations ×7



Current observation

Frozen ViT + trainable multimodal projector

Q-identity
16 tokens / past frame

Cached past boxes

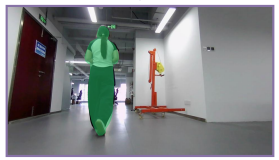
Current-frame tokens
256 tokens



DobotTrack VLA language model

Instruction
History (7 × 16)
Current (256)
Mask
BBox
Trajectory queries ×8

SAM2 decoder



Grounding head



Action head



(b) Q-identity

Projected visual tokens Xτ
256 × d

Input projection
+ 2D position embedding

Full-image features Fτ

Cached target box bτ

BBox patch selection

Target context Tτ
(or no-target token)

16 learned queries

×3

Concat [Qτ ; Tτ]

Target-conditioned self-attention

×16

K, V

Full-image cross-attention

Feed-forward network

LayerNorm + output projection

Compressed tokens Cτ (16 × d)